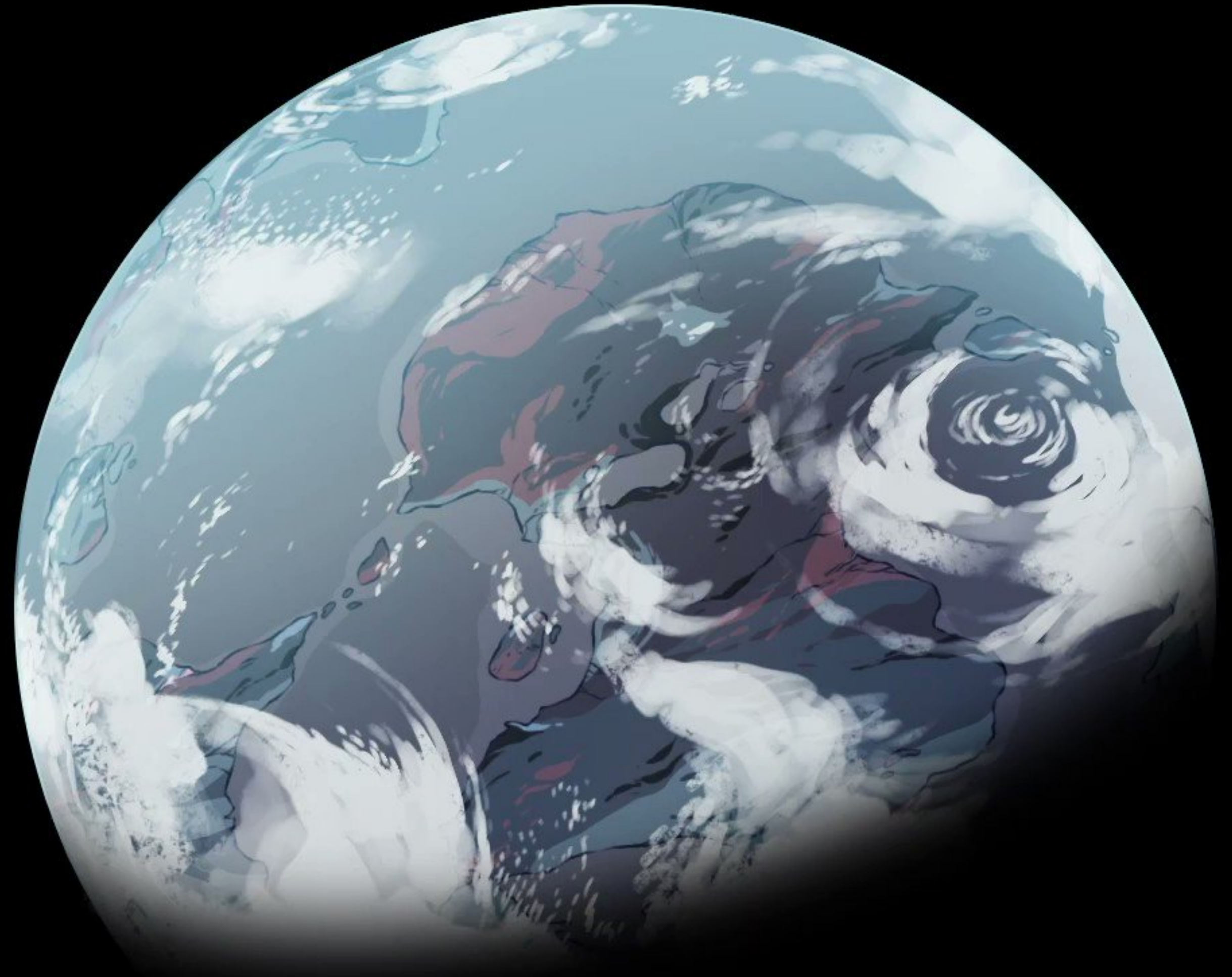


Science & Science Fiction

33-120 | Prof. Katie Breivik



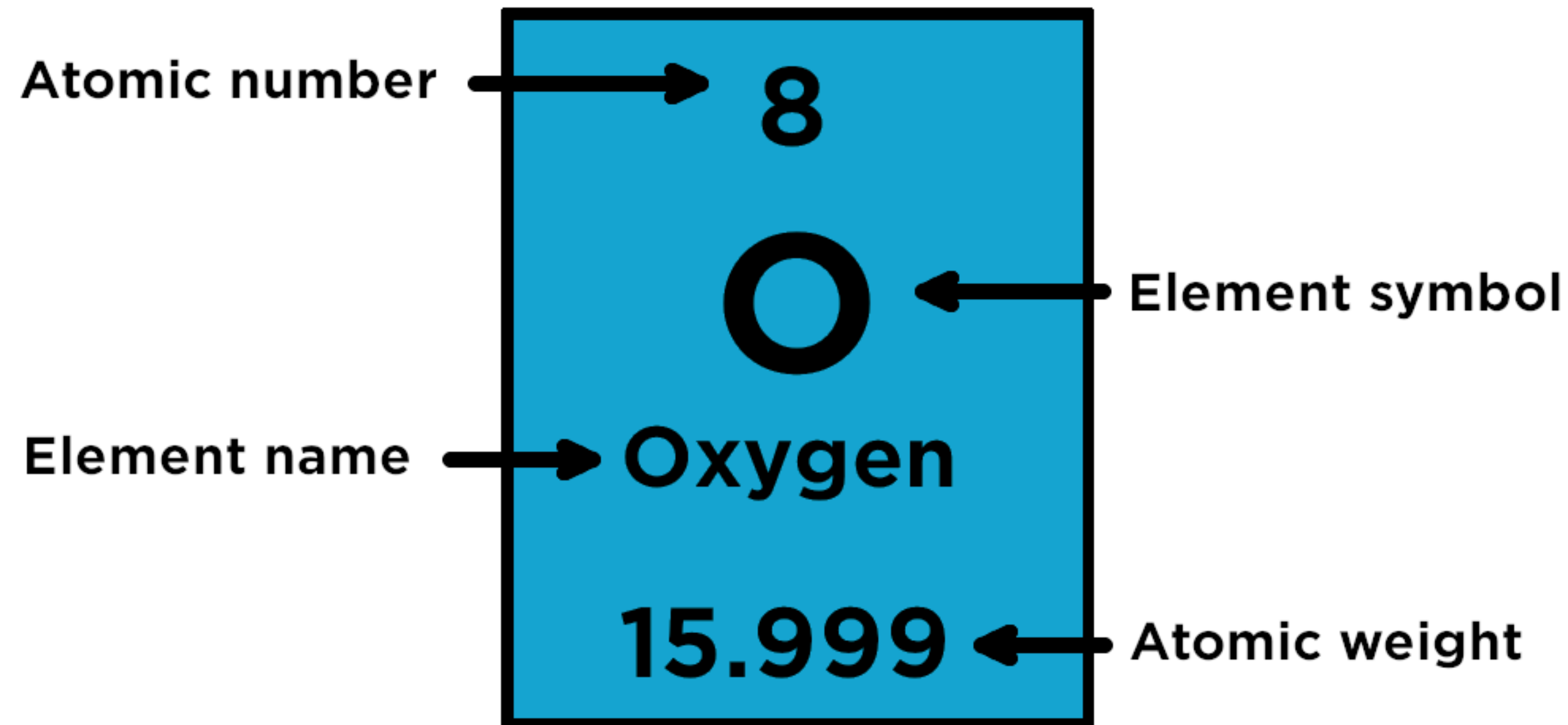
Announcements

- ▶ Reminder: please send me an email by the end of today if you'd like to engage with media beyond the listed films on the Film Critique assignment!
- ▶ Slight shift in Problem 4 schedule:

Previously: available Feb 14, due Feb 21

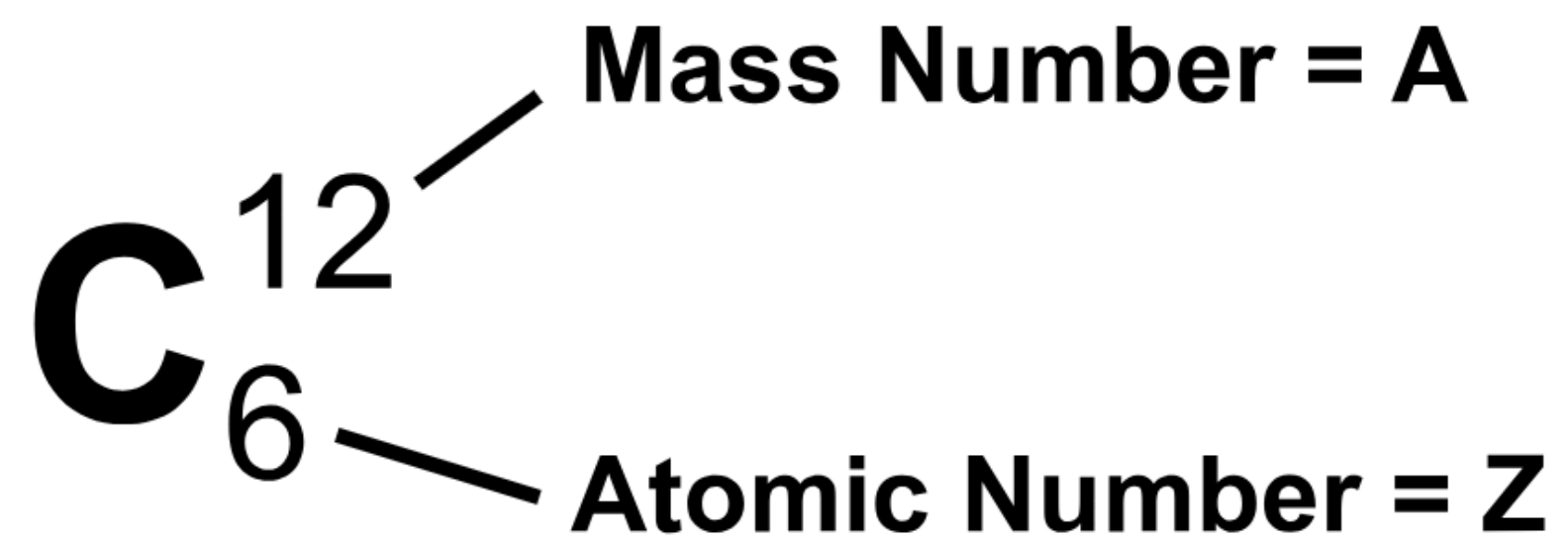
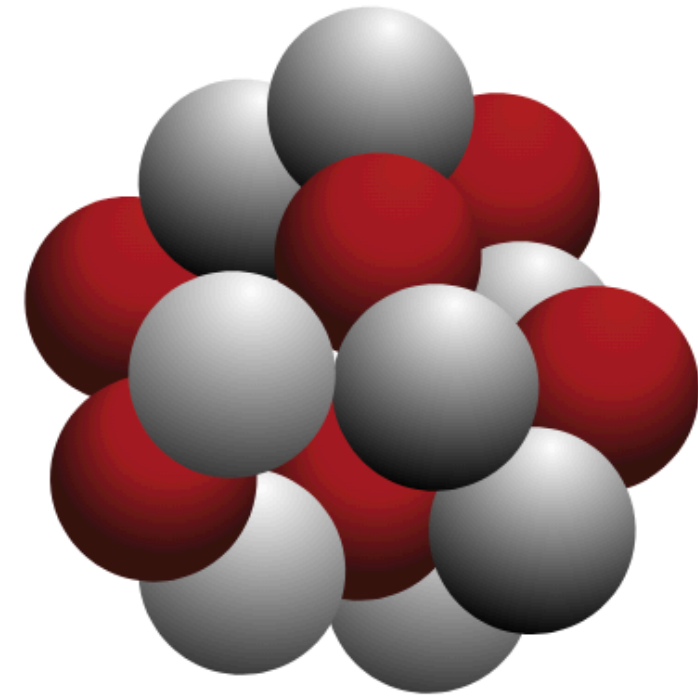
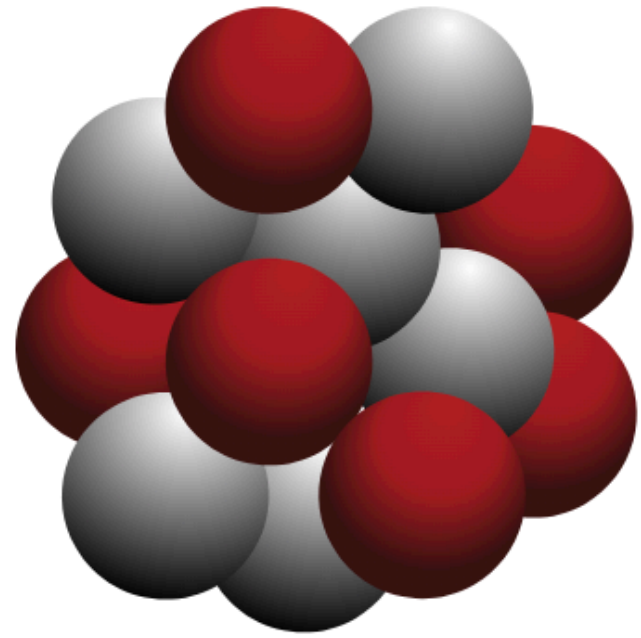
Now: available Feb 17, due Feb 26

atoms: collections of protons, neutrons, and electrons



Atomic number:
of protons

Atomic weight:
average mass of
the atom
accounting for
the relative
abundance of
each *isotope*



Isotopes:

chemical variants of atoms with the same # of protons but *different # of neutrons*

Mass number: $A = n + p$



Forbidden Planet

Directed:
Fred McLeod Willcox

MGM (1956)



- Common lead would have crushed the vehicle sir.
- This is my morning's run of isotope 217.
- The whole thing hardly comes to ten tons.

—Robby the Robot

Common lead would have crushed the vehicle sir.

—> What is “common lead”?

—> And what are the other types?

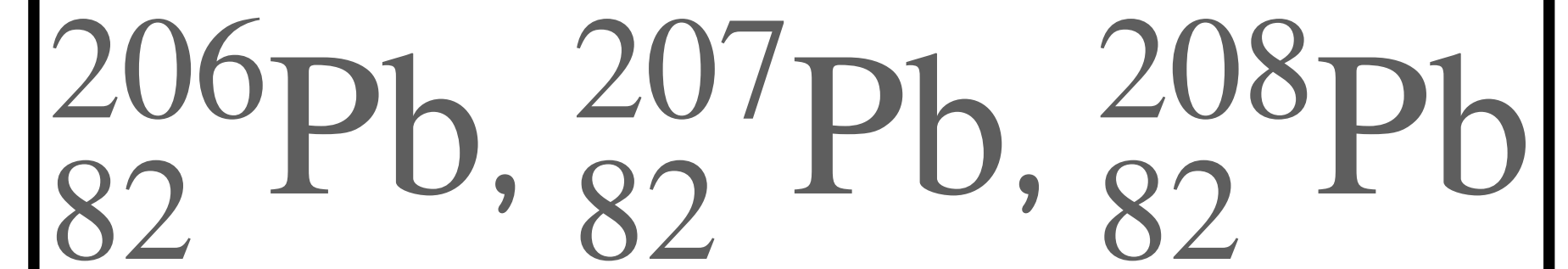
This is my morning’s run of isotope 217.

—> Is there something special about isotope 217?

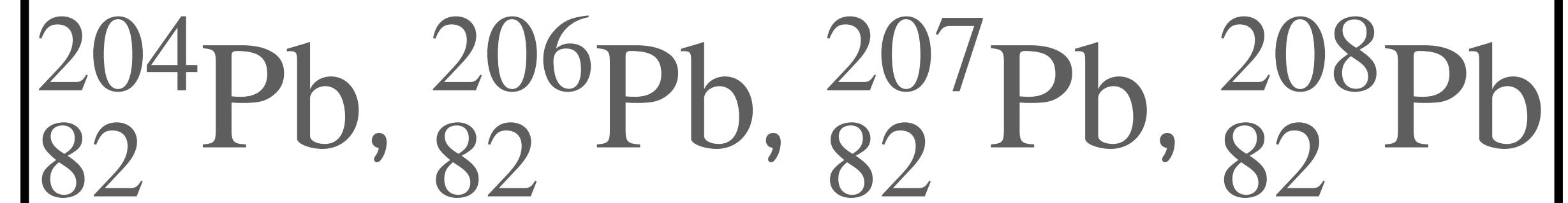
The whole thing hardly comes to ten tons.

—> Should isotope 217 be lighter than common lead?

There are three isotopes of common lead:



There are four stable isotopes of lead:



There are *dozens* of non-stable isotopes of lead:



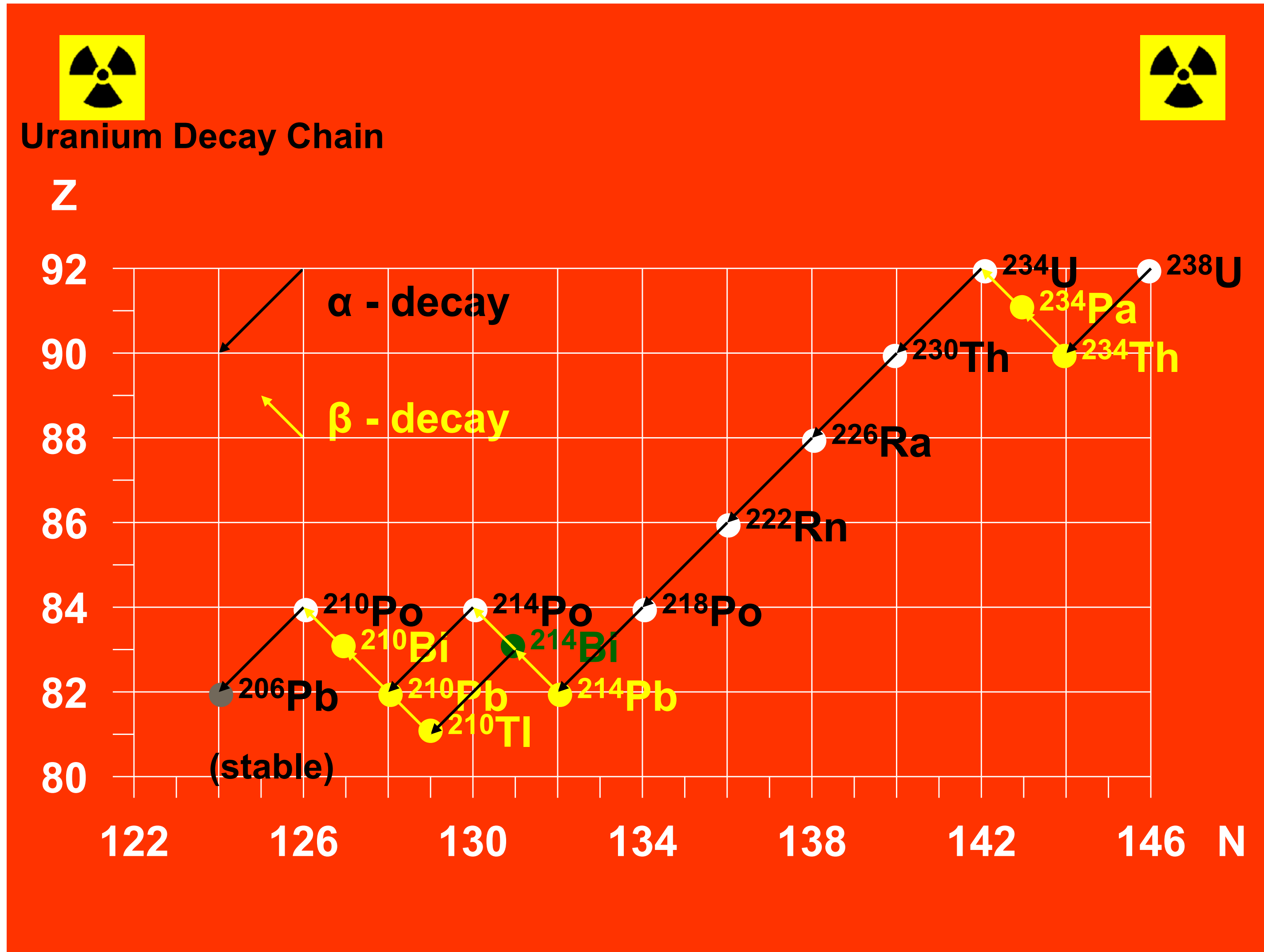
—> Is there something special about isotope 217?

$^{217}_{82}\text{Pb}$ has 10 extra neutrons compared to $^{207}_{82}\text{Pb}$

—> Should isotope 217 be lighter than common lead?

10 extra neutrons works out to 5% extra mass —> heavier!

$^{217}_{82}\text{Pb}$ decays after 20s —> wouldn't be $^{217}_{82}\text{Pb}$ at the end of the clip!



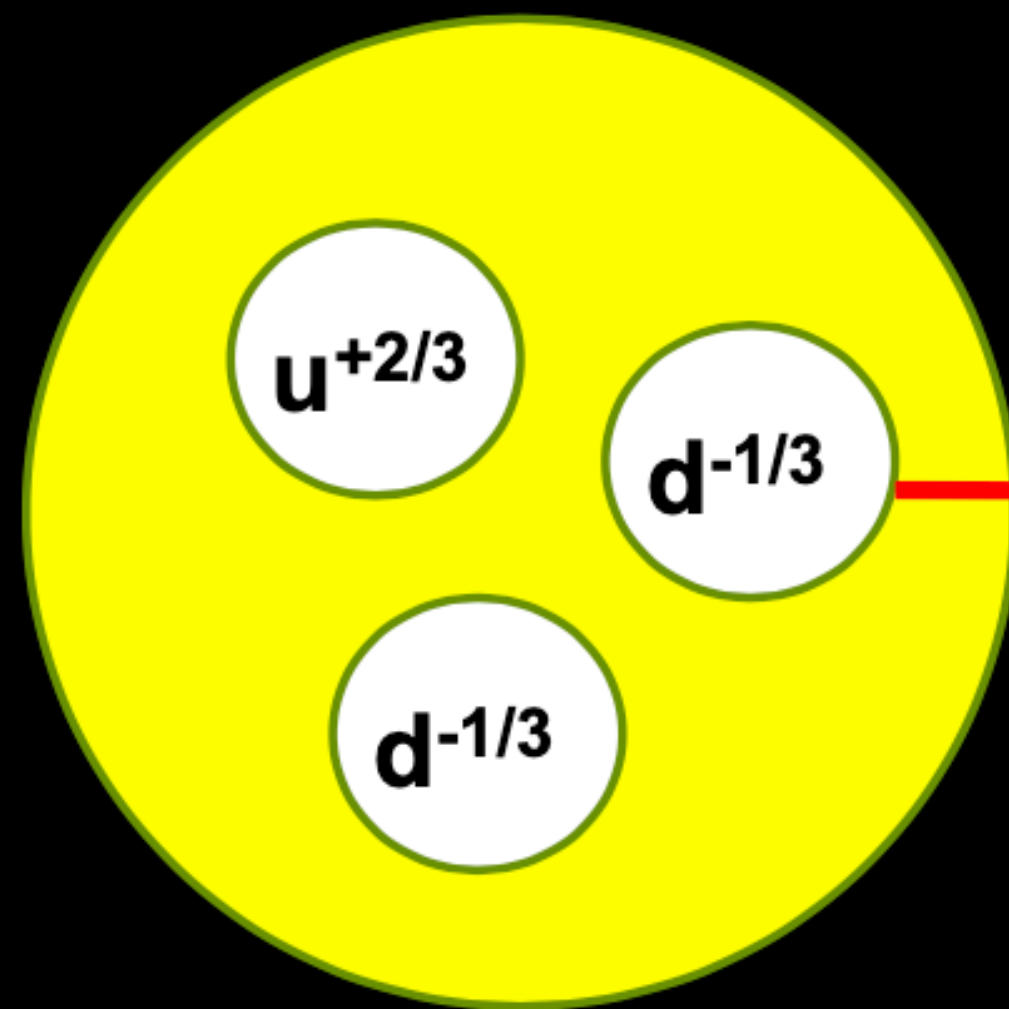
β -decay: atom emits a
“ β particle”
(an energetic e^- or e^+)

maintains the mass
number but changes
the atomic number

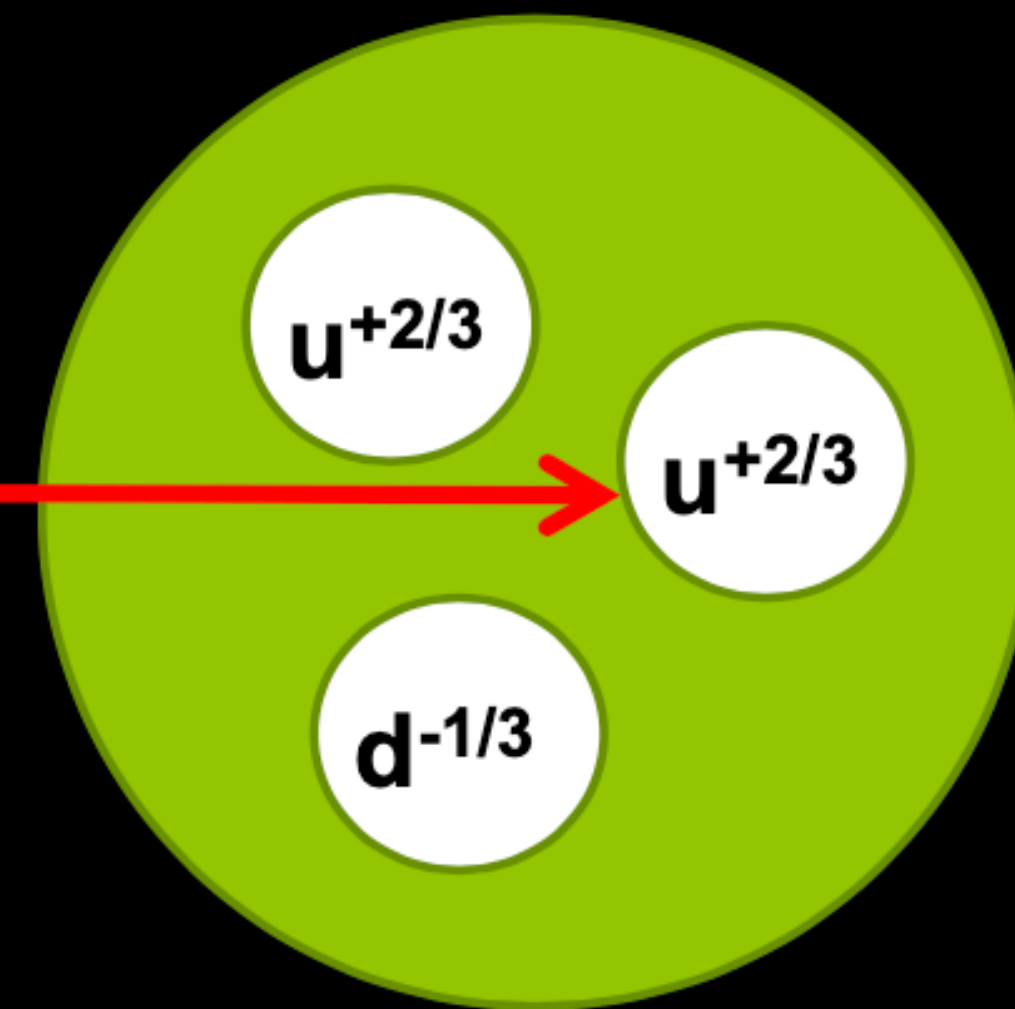
A consequence of the
weak nuclear force

Beta Decay

Neutron (charge: 0)



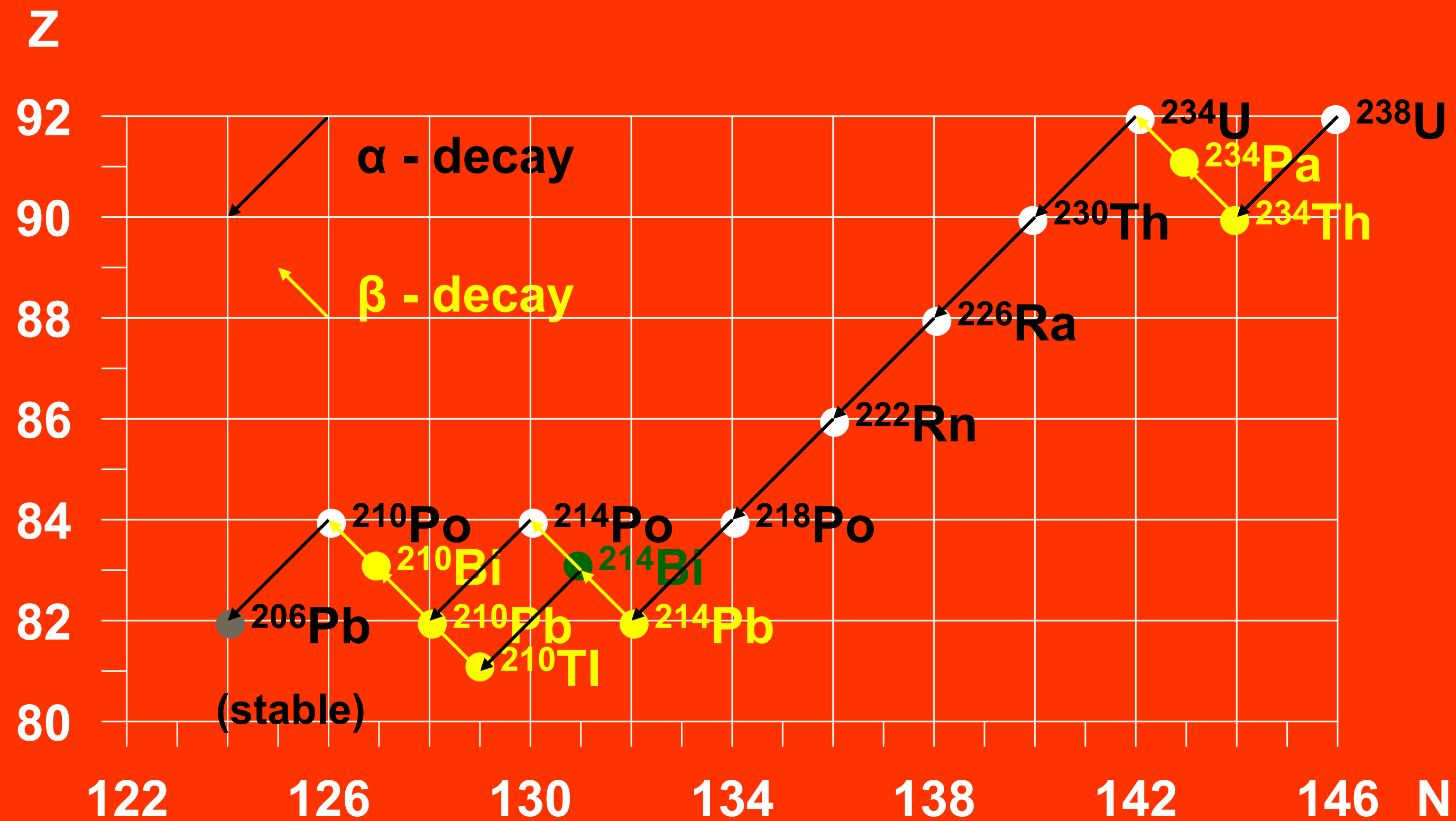
Proton (charge: +1)



$d^{-1/3}$ decays to a $u^{+2/3}$ and a W^- boson



Uranium Decay Chain



α -decay: atom emits an “ α particle” (helium nucleus)

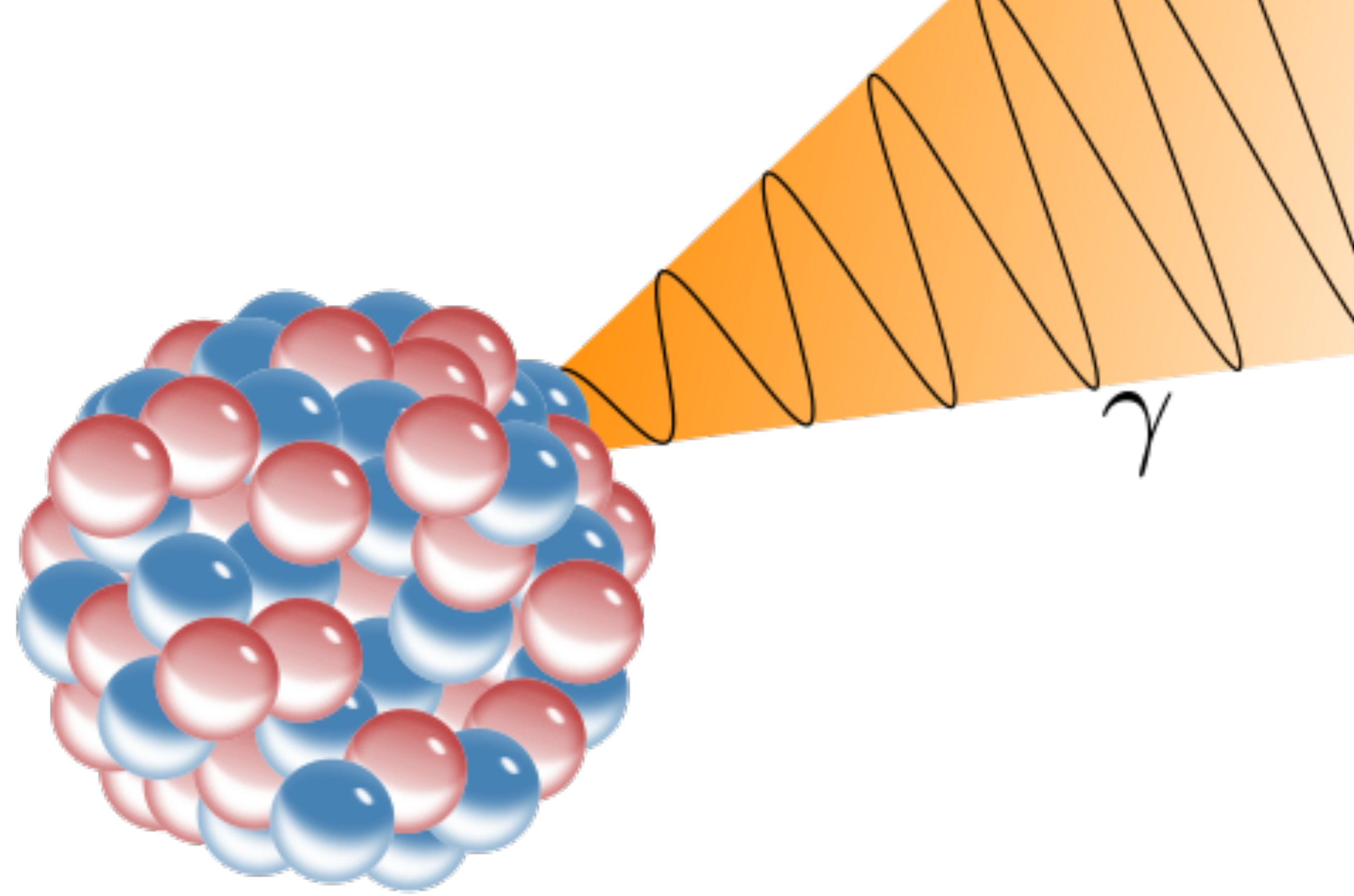
decreases the mass number by 4 and the atomic number by 2

A consequence of the *strong* nuclear force

γ decay is the emission
of a γ -ray photon

an excited e- will decay
to a ground state and
emit the photon

happens due to a
combination of the
electromagnetic/strong
nuclear force



α -decay

β -decay

γ -decay

changes mass

changes charge

changes energy

emits He

emits e^- or e^+

emits γ -ray

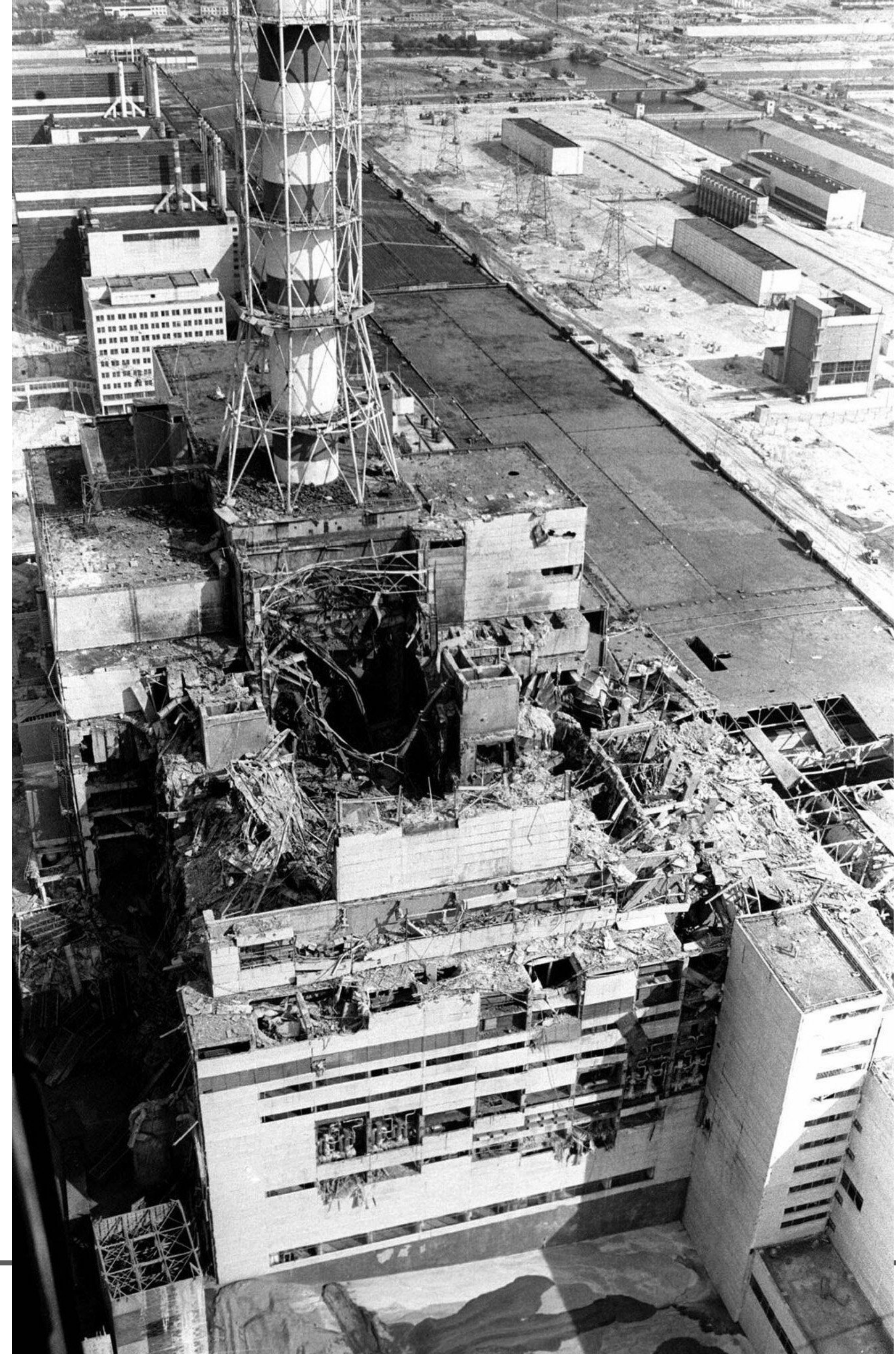
dangerous at
tiny scales

dangerous at
~inch scales

dangerous at
~hundred yard scales

People affected by the Chernobyl disaster were exposed to *all* kinds of radioactive decay

Today, α -decay and β -decay are still sources of radiation within the exclusion zone





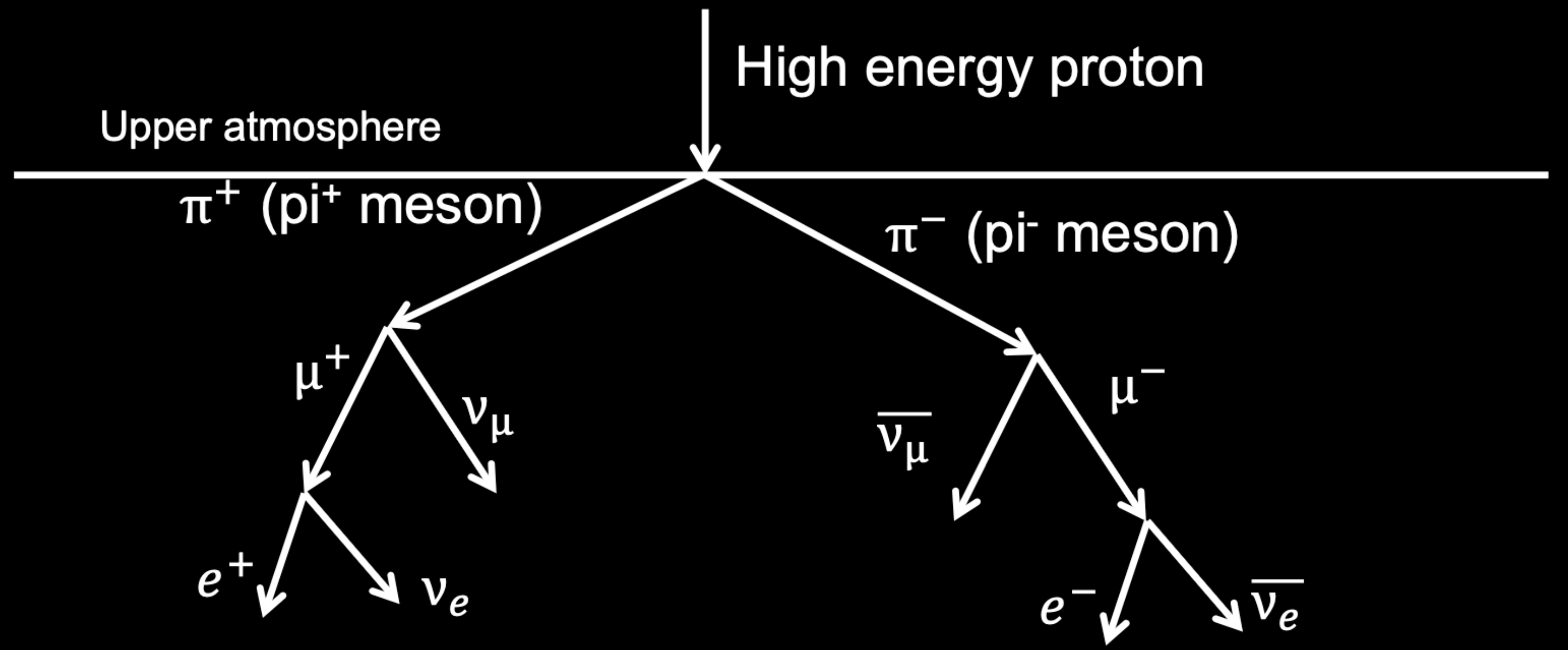
Chernobyl

Written:
Craig Mazin

HBO (2019)

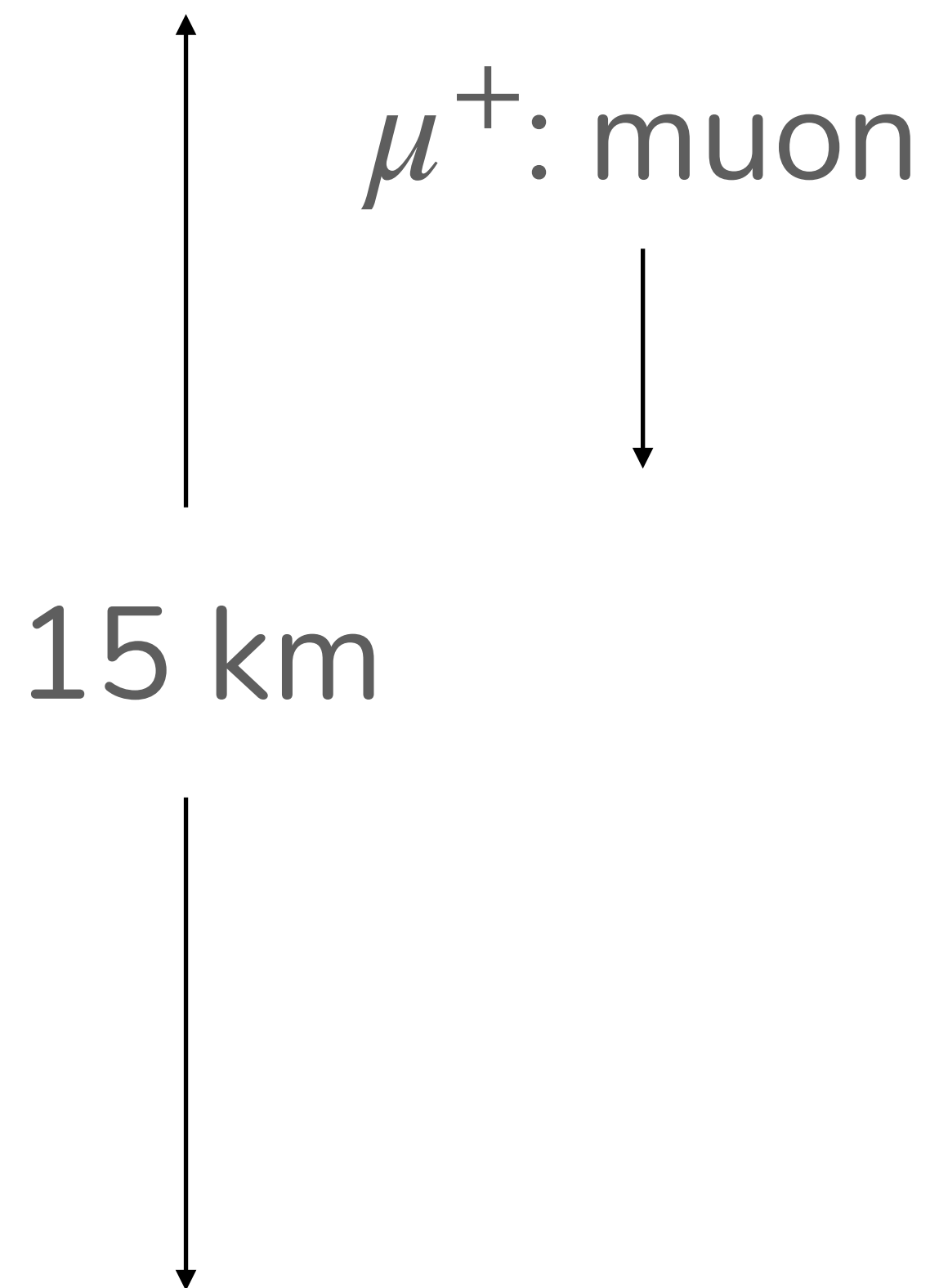
combining nuclear physics & special relativity

Production of Cosmic Ray Muons



combining nuclear physics & special relativity

upper atmosphere



μ^+ lifetime is 2.2 microseconds

The flight time at the speed of light is:

$$\Delta t = \Delta x / c = (15 \times 10^3 \text{ m}) / (3 \times 10^8 \text{ m/s}) = 5 \times 10^{-5} \text{ s}$$

50 microseconds > 2.2 microseconds!

combining nuclear physics & special relativity

upper atmosphere

μ^+ : muon

15 km

Time dilation at $v/c = 0.9995$

$$\Delta t = \frac{\Delta t'}{\sqrt{1 - v^2/c^2}} = 31.6 \Delta t'$$

The flight time at the speed of light is:

$$\Delta x = \Delta t \times 0.9995c$$

$$\Delta x = 31.6 \times 2.2 \times 10^{-6} \text{ s} \times 0.9995 \times 3 \times 10^8 \text{ m/s} = 20.9 \text{ km}$$

Next time:

- ▶ Atoms and the kinetic theory of gases

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